

From Fractional to Rough Volatility: Forecasting Performance of the Comte–Renault, RFSV and Rough Bergomi Models

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Table of Contents

► Setup

► Data & Methods

► Results

► Conclusion & Future Work



Motivation & Research Question

1 Setup

Volatility is a central concept in modern quantitative finance, influencing derivative pricing, risk management, and portfolio allocation. Finding accurate ways to model volatility has proved to be a difficult and ongoing challenge, making volatility a natural choice for this thesis.

Research Question

Does the greater complexity of rough volatility models lead to measurable improvements in forecasting volatility compared with more traditional long-memory models?



Models at a Glance

1 Setup

Comte-Renault (CR) (fOU, $H > \frac{1}{2}$) — long memory

$$dX_t = -k X_t dt + \nu dW_t^H, \quad X_0 = 0, k > 0, H \in (\frac{1}{2}, 1)$$

RFSV ($H < \frac{1}{2}$) — rough short-run dynamics

$$dX_t = \nu dW_t^H - \alpha (X_t - m) dt, \quad X_t = \log \sigma_t, H < \frac{1}{2}$$

Rough Bergomi (rBergomi) ($H < \frac{1}{2}$) — rough, forward variance

$$V_t = \xi_0(t) \exp\left\{\eta W_t^H - \frac{1}{2}\eta^2 \text{Var}(W_t^H)\right\}$$

Benchmark: HAR-RV (D/W/M) $\ell_{t+1} = \beta_0 + \beta_d \ell_t + \beta_w \ell_t^{(w)} + \beta_m \ell_t^{(m)} + \varepsilon_{t+1}$



Table of Contents

► Setup

► Data & Methods

► Results

► Conclusion & Future Work



Dataset & Target

2 Data & Methods

- **Data:** Realized Library developed by the Oxford-Man Institute, providing daily measures of realized variance from 2000 to 2018; 6 indices ($\sim 4,000$ trading days).
- **Split:** 70% train, 15% validation, 15% test.
- **Target:** One-day-ahead log-volatility $\Rightarrow \ell_{t+1} = \frac{1}{2} \log(\text{RV}_{t+1})$.
- **Metrics:** **QLIKE** (on variance), **MSE-log**.

QLIKE

$$L_{t+1} = \frac{\text{RV}_{t+1}}{\widehat{\text{V}}_{t+1}} - \log\left(\frac{\text{RV}_{t+1}}{\widehat{\text{V}}_{t+1}}\right) - 1$$



Implementation Pipeline

2 Data & Methods

- **Step 1 — Data preprocessing:** cleaning, alignment, and construction of HAR-type features (daily, weekly, and monthly averages of log-RV).
- **Step 2 — Model calibration:** estimation of each model's **parameters** by minimizing the validation QLIKE.
 - **Comte-Renault:** grid search on (H, K) .
 - **RFSV:** H estimated via **RKS**; (K, α) selected by minimizing validation QLIKE.
 - **Rough Bergomi:** H fixed (RKS), grid over (K, α, η) on validation set.
- **Step 3 — Forecasting & evaluation:** one-day-ahead predictions evaluated through **test QLIKE** and **rolling-window QLIKE**, ensuring a robust evaluation of each model's performance.



Estimated Roughness (RKS)

2 Data & Methods

RKS (Bianchi & Angelini, 2025): compares log-volatility changes across time scales using a Kolmogorov–Smirnov distance.

It uses the **full distribution** of log-volatility increments, providing a more **robust and reliable** estimate of H .

All indices show $H < 0.5 \rightarrow$ volatility is **rough**.

Index	H
Euro Stoxx 50	0.17
FTSE 100	0.12
S&P 500	0.14
Nasdaq Comp.	0.15
Nikkei 225	0.14
Hang Seng	0.08



Table of Contents

► Setup

► Data & Methods

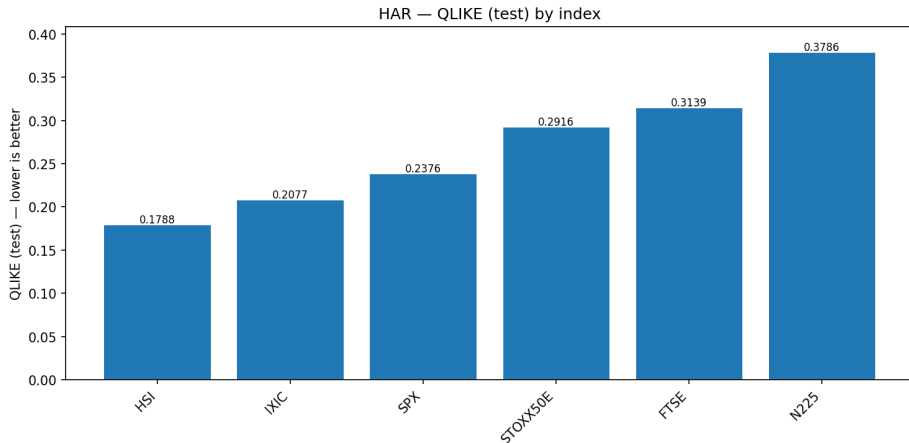
► **Results**

► Conclusion & Future Work



HAR Benchmark & Baseline

3 Results

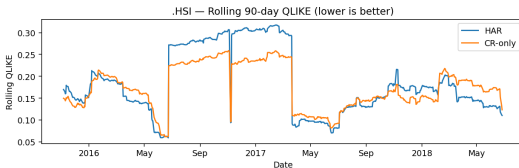
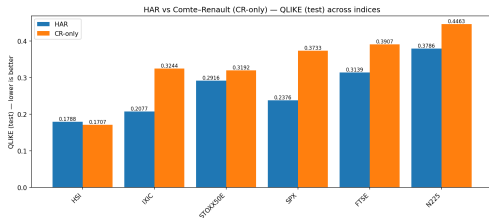


QLIKE (test) — lower is better — by index



CR vs HAR — Test & Rolling

3 Results



Bar chart (left): HAR outperforms CR on most indices, with HSI as the only clear exception. No systematic long-memory advantage.

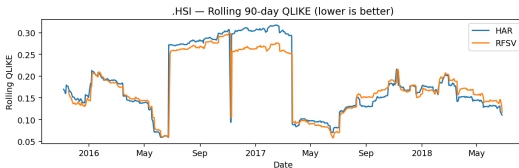
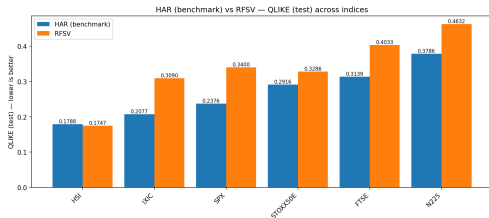
Rolling HSI (right): CR performs better in persistent phases but reacts slower to volatility spikes.

Summary: CR helps when persistence dominates, yet average gains remain modest.



RFSV vs HAR — Test & Rolling

3 Results



Bar chart (left): RFSV delivers similar performance to HAR across indices, with HSI being the only clear case of improvement. No consistent benefit from roughness.

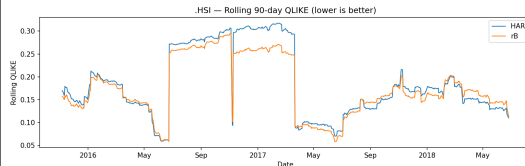
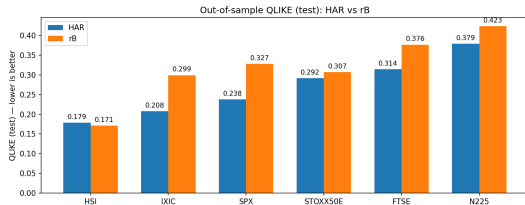
Rolling HSI (right): RFSV performs slightly better in stable periods, while HAR reacts faster to volatility spikes.

Summary: Roughness can help in specific regimes, but overall forecast gains remain modest.



rBergomi vs HAR — Test & Rolling

3 Results



Bar chart (left): rBergomi performs similarly to HAR across indices, with HSI showing the clearest improvement.

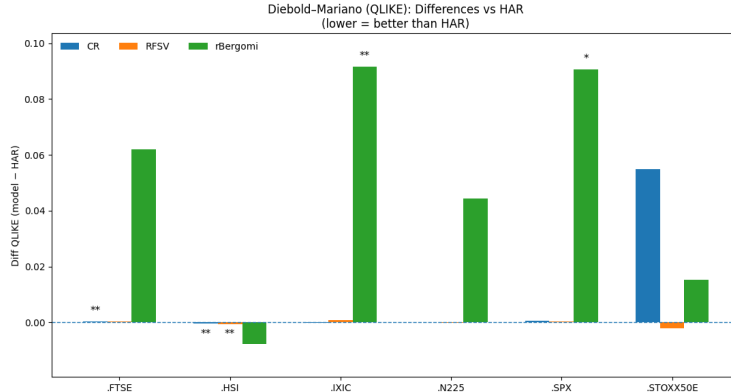
Rolling HSI (right): rBergomi similar to HAR in calm times, better during volatility spikes.

Summary: rBergomi offers local gains, but no systematic improvement over HAR.



Diebold-Mariano (QLIKE) — Differences vs HAR

3 Results



CR and RFSV significantly outperform HAR only on HSI, while rBergomi shows significant losses on the Nasdaq Composite (IXIC) and the S&P 500 (SPX).



Table of Contents

► Setup

► Data & Methods

► Results

► Conclusion & Future Work



Contextualizing the Results

4 Conclusion & Future Work

- The findings must be **interpreted within the daily-frequency setting** adopted in this thesis.
- Using **daily data** smooths the roughness that becomes evident at higher frequencies.
- The superior performance of **HAR-RV** in five out of six indices reflects its **better alignment with the scale and information content** of daily data.

HAR-RV, though simple, perfectly matches the scale and informational content of daily data, making it the most effective model for out-of-sample forecasting.



Limitations & Future Work

4 Conclusion & Future Work

- The implementations of fractional and rough models are **simplified and discrete**, with **fixed parameters** calibrated on historical data.
- These design choices ensure strict out-of-sample comparability but may limit the models' adaptability to changing volatility regimes.
- Future research should explore **dynamic re-estimation**, high-frequency data, and continuous-time calibrations to better capture roughness.



From Fractional to Rough Volatility: Forecasting Performance of the Comte–Renault, RFSV and Rough Bergomi Models *Thank you for listening!*

Any questions?